



$$Z = \frac{600}{200 \times 10^{-3}} \Rightarrow 3k\Omega$$

Inductance of meter coil for full scale deflection

$L =$

$$\frac{(0.1 + 0.2 \times 100 \times \pi / 180)}{4\pi} \Rightarrow 0.0357$$

Reactance of meter coil at 50Hz

$$= 2\pi \times 50 \times 0.0357 \Rightarrow 11.22 \Omega$$

$\therefore$ Series resistance required

$$\begin{aligned} R_s &= \sqrt{(3000)^2 - (11.22)^2} \\ &= 2999.9 \\ &\approx 3k\Omega \end{aligned}$$

Example 9:

Which type of the instrument has the high power consumption?

- (a) Moving iron
- (b) PMMC
- (c) Hotwire
- (d) All instruments have identical power consumption

Ans: (c)

Errors in M.C & M.I Instruments

1. Frictional Error

To reduce frictional error, so for the same developed torque, weight of the system should be

less than torque i.e., $\left(\frac{\text{torque}}{\text{weight}}\right)$ ratio has to be high

In MC Instrument coil is moving hence weight of moving system is more compared to MI

Note: If weight decreases, Torque/ weight increases sensitivity increases, error ↓, Accuracy ↑

Frictional error is moderately more in M.C compared to M.I

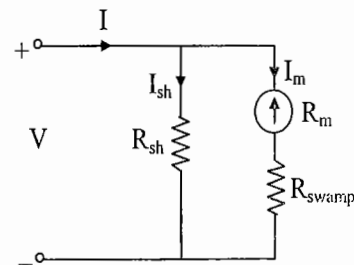
2. Temperature Error.

- Because of temperature spring is expand and indicates less than the actual value and coil resistance also changes
- D.C Current is present in MC and RMS current is present in M2, Hence the heat developed is more for RMS current compared to D C Current
- Temperature error is more in MI than MC

In MC instrument

The temperature effect is more in Ammeters than voltmeters because ammeter resistance is very less.

To reduce temperature effect in Ammeter a 'swamping resistor' is connected in series with meter.



$$I_m = \frac{I R_{sh}}{R_{sh} + R_m + R_{swamp}}$$

$I/I_m = m = \text{multiplying factor}$

$$\frac{R_{sh} + R_m + R_{swamp}}{R_{sh}} = \frac{I}{I_m} = m$$



$$1 + \frac{R_m + R_{\text{swamp}}}{R_{\text{sh}}} = m$$

$$R_{\text{sh}} = \frac{(R_{\text{swamp}} + R_m)}{m - 1}$$

R_{swamp} is made of manganin for D.C

R_{swamp} is made of constantan for A.C

Its Value is 30-40 times the ammeter resistance.

The value of shunt resistance (with swamping resistor) is more than the value of shunt resistance (without swamping resistor)

Conclusion:

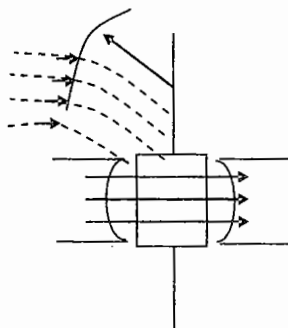
To reduce temperature effect swamping resistor is used

3. Frequency error

If same Input is measured at different frequencies then the meter gives different values, then it is called frequency error.

- MC used for DC so no frequency error in MC
- MI measures AC hence frequency error is present

Stray magnetic Field error



MC → Paramagnetic is used (strong magnetic field)

M.I → electromagnetic (weak magnetic field)

- Due to external magnetic field Inside meter operating field is disturbed. It is called stray magnetic field error.
- Stray magnetic field depends on magnitude of external magnetic field and its direction
- when magnitude of external magnetic field is in phase with main field, stray magnetic field error will be more
- To reduce this error Iron shielding is used
- The stray magnetic field error is more in MI compared to MC

Hysteresis & Eddy current Error in M.I

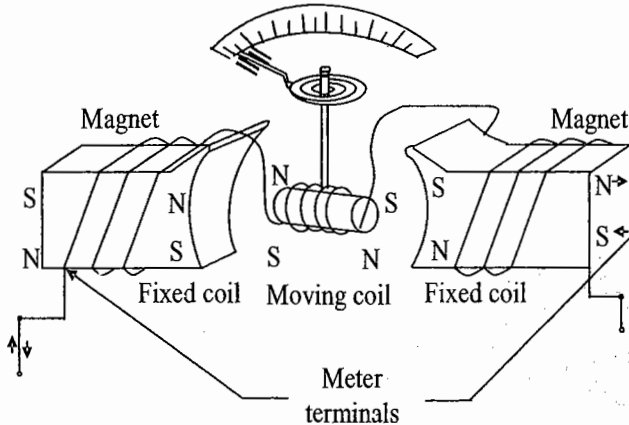
Hysteresis Error:

- Because the use of high permeability soft iron has large hysteresis loss than this error will be more compare to MC
- To reduce the hysteresis loss Nickel-Iron alloy is used In the manufacturing of M.I which has low value of hysteresis area & hence minimum error

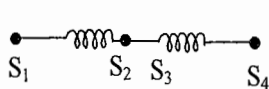
Eddy Current Error :

- In MI Measurement eddy current error is $\propto f^2$
- Eddy current error is constant for high frequency. Hence because of this reason M.I Instrument is suitable up to 125 Hz only
- Compared to M.C temperature $\uparrow$, stray load losses $\uparrow$, Hysteresis frictional loss $\downarrow$.
- To measure very low currents & voltage at very high frequency (50 MHz) thermocouple ammeter and thermocouple voltmeters are used

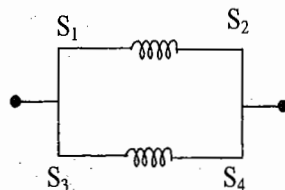
Electro Dynamometer



- In dynamometer type instruments air core is used for both fixed coil and moving coil. Hence this instrument is almost free from hysteresis and eddy current error. Hence it is used as transfer type instrument for power frequency
- Thermocouple instrument can be used as transfer type instrument for high frequency.



Series for low currents



Parallel for high currents

Fixed coil connections for low and high currents

Operation:

It works on principle of dc motor ie when a current carrying conductor placed in a magnetic field experiences a mechanical force.

$$T_d = I_1 I_2 \cos \phi \frac{dM}{d\theta}$$

Note: $\cos \phi$ here is not a power factor

I_1 = fixed/current coil current

I_2 = moving/potential coil current

θ = angle between I_1 and I_2

M = Mutual inductance

- Spring control is used to produce controlling torque (springs are carrying current)

$$T_c = k_c \theta$$

- Air friction damping is used also called **pneumatic damping**

At final steady state position $T_c = T_d$

$$\theta = \frac{I_1 I_2 \cos \phi}{k_c} \frac{dM}{d\theta}$$

$$\text{For dc: } T_d = I_1 I_2 \frac{dM}{d\theta}$$

$$\text{If } I_1 \text{ \& } I_2 \text{ are same then } T_d = I^2 \frac{dM}{d\theta}$$

EMMC used as

Voltmeter, Ammeter and wattmeter for ac and dc.

Wattmeter is more popularly used.

Application

To convert as dynamometer ammeter connect both fixed coil and moving coil in series. This combination in series with load

	PMMC	MI	EMMC/Dynamometer
T_d	BINA	$\frac{1}{2} I^2 \frac{dL}{d\theta}$	$I_1 I_2 \cos \phi \frac{dM}{d\theta}$
θ	$\frac{BINA}{k_c}$	$\frac{1}{2} \frac{I^2}{k_c} \frac{dL}{d\theta}$	$\frac{I_1 I_2 \cos \phi}{k_c} \frac{dM}{d\theta}$



Example 10:

The mutual inductance of a dynamometer ammeter varies with deflection θ (degree) as $M=6\cos(\theta+30^\circ)$ mH.

The deflection torque produced by a current of 50 mA corresponding to a deflection of 60° is

- (a) $10 \mu\text{N-m}$ (b) $15 \mu\text{N-m}$
(c) $20 \mu\text{N-m}$ (d) $25 \mu\text{N-m}$

Ans: (b)

Sol: $T_d = I^2 \frac{dM}{d\theta};$

$$M = 6 \cos(\theta+30^\circ) \times 10^{-3}$$

$$\left. \frac{dM}{d\theta} \right|_{60^\circ} = 6 \sin(\theta+30^\circ) \times 10^{-3}$$

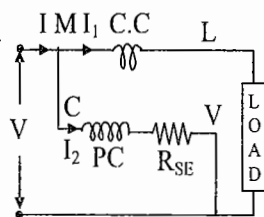
$$= 6 \times 1 \times 10^{-3} = 6 \times 10^{-3}$$

$$T_d = (50 \times 10^{-3})^2 \times 6 \times 10^{-3}$$

$$= 15000 \times 10^{-9} = 15 \mu\text{N-m}$$

• **To convert as Dynamometer wattmeter:**

Connect current coil in series with the load and potential coil in parallel with load



M – Main

$I_1 = I$

L – Load

$$I_2 = \frac{V}{R_s}$$

C – common

V – voltage

$$T_d = I_1 I_2 \cos \phi \frac{dM}{d\theta}$$

$$= I \frac{V}{R_s} \cos \phi \frac{dM}{d\theta}$$

$$= VI \cos \phi \left(\frac{1}{R_s} \frac{dM}{d\theta} \right)$$

$$T_d = P \left(\frac{1}{R_s} \frac{dM}{d\phi} \right)$$

$$\text{Power} = VI \cos \phi$$

At balance

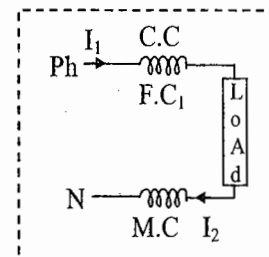
$$T_c = T_d$$

$$\theta = \frac{P}{R_s} \frac{dM}{d\theta}$$

$$\theta \propto P$$

It is average active power.

• **To convert as dynamometer ammeter.**



FC – current coil

MC – voltage coil (or) potential coil

$$I_d = I_1 = I_2$$

$$T_d = I^2 \cos \phi \frac{dM}{d\theta}$$

$$T_c = k_c \theta$$

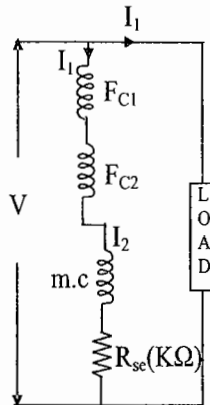
$$k_c \theta = I^2 \cos \phi \frac{dM}{d\theta}$$

$$\theta \propto I^2$$



Non linear scale

- This type ammeter when used for ac reads rms value.
- **To convert as dynamometer voltmeter**
Fixed coil and moving coil connected in series and this combination is connected in parallel with the load.



$$I_1 = I_2 = \frac{V}{R_{sc}}$$

$$T_d \propto I^2$$

$$T_d \propto \frac{V^2}{R_{sc}^2}$$

$$T_c = k_c \theta$$

$$T_c = T_d$$

$$\theta = \frac{V^2}{R_{sc}^2 k_c} \frac{dM}{d\theta}$$

$$\theta \propto V^2$$

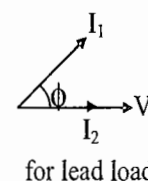
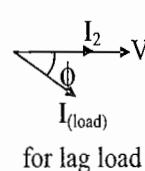
- Non – linear scale used for ac and dc measurements
- Moving coil measures potential hence it's called potential coil
- Fixed coil carries the load current and it is connected in series with the load hence it's called current coil.

Error due to type of wattmeter connection:

MC connection	LC connection
(MC short dia)	(LC Short dia)
Measured total power = $P_m = P_{load} + I^2 r$ $P_m = P_T + I^2 r$ Error = $\left(\frac{P_m - P_T}{P_T} \right) \times 100$ $\text{Error} = P_m - P_T = I_1^2 r$ $r \rightarrow$ Resistance of current coil	Assume CC resistance is small Measured power $= p_m = P_{load} + \frac{V^2}{R_s}$ $\text{Error} = P_m - P_{load}$ $= \frac{V^2}{R_s} = I_2^2 R_s$ $I_2 = I - I_1$
Error is minimum if load current 'I' is small	For large load current $I_1 \uparrow, I_2 \downarrow = I - I_1 \uparrow$ I_2 is small, so error $I_2^2 R_s$ decreases Error is minimum if load current is high

$$I^2 r = \frac{V^2}{R_s}$$

- A potential coil is connected with a series resistance (R_s) which is in the order of $K\Omega$ hence the inductive reactance of the PC is small and neglected.
- Current flowing through potential coil is in phase with voltage 'V'



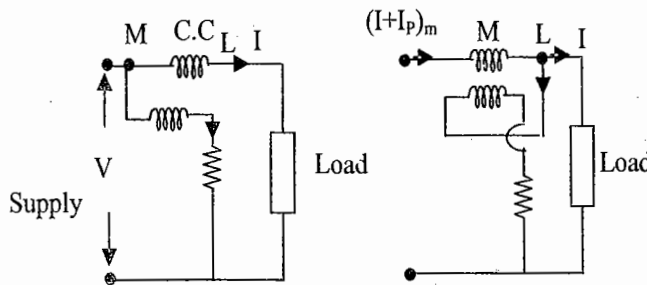


Note: Thickness of the current coil conductor is higher to withstand the higher load currents

- Standard conductors are used to reduce the eddy current losses.

Example 11:

A 250V, 10A electrodynamic wattmeter has a resistance of current and potential coil of 0.5Ω and 12500Ω respectively. Find the percentage error due to each of the two methods of connections shown in figure below with unity power factor loads at 250V. With currents 12A neglect error due to pressure coil inductance.



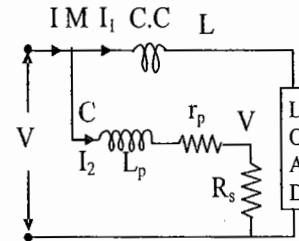
- (a) 2.54% ; 0.146% (b) 2.34% ; 0.166%
(c) 2.43% ; 0.616% (d) 2.15% ; 0.158%

Ans: (b)

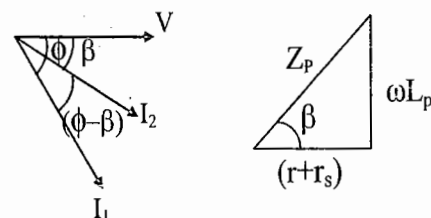
$$\begin{aligned} \text{Sol: } P_t &= V_L I_L \cos \phi & P_t &= V_L I_L \cos \phi \\ &= 250 \times 12 \times 1 & &= 250 \times 12 \times 1 \\ &= 3000 \text{ W} & &= 3000 \text{ W} \\ (P_m)_{MC} &= P_t + I_L^2 R_C & (P_m)_{LC} &= P_t + \frac{V^2}{R_p} \\ &= 3000 + (12)^2 \times 0.5 & & \\ &= 3000 + \frac{(250)^2}{12500} & & \\ \% \text{ error} &= \frac{3072 - 3000}{3000} \times 100 & & \\ &= \frac{3005 - 3000}{3000} \times 100 & & \\ &= 2.4\% & & = 0.166\% \end{aligned}$$

Error in Watt Meters

Error due to potential coil inductance:



L_p = inductance of PC
 r_p = resistance of PC ($r_p \ll R_s$)
 R_s = series multiplier resistance ($K\Omega$)
 Z_p = impedance of PC
 $= ((r_p + r_s) + j\omega L_p)$
 β = PC phase angle



$$\begin{aligned} \cos \beta &= \left(\frac{R_s}{Z_p} \right) \\ \beta &= \tan^{-1} \left(\frac{\omega L_p}{r_p + r_s} \right) \end{aligned}$$

$$T_d \propto I_1 I_2 \cos(\phi - \beta) \frac{dM}{d\theta}$$

W_M (measured power) $\propto \theta$

$$= \frac{V}{Z_p} \frac{I}{K_c} \cos(\phi - \beta) \frac{dM}{d\theta}$$

$$W_M = \frac{VI \cos \beta \cos(\phi - \beta)}{R_s K_c} \frac{dM}{d\theta}$$

W_M (True power) without ' L_p '



$$T_d \propto I_1 I_2 \cos \phi \frac{dM}{d\theta}$$

$$T_d \propto \frac{VI \cos \phi}{R_s} \cdot \frac{dM}{d\theta}$$

$$W_T = \frac{VI}{R_s} \cos \phi \frac{dM}{d\theta}$$

$$\frac{W_T}{W_m} = \frac{\cos \phi}{\cos \beta \cos(\phi - \beta)}$$

$$\text{Correction factor (c.f.)} = \frac{\cos \phi}{\cos \beta \cdot \cos(\phi \mp \beta)}$$

'-' for lag

'+' for lead

$$W_T = \frac{\cos \phi}{\cos \beta (\cos \phi \cdot \cos \beta + \sin \phi \cdot \sin \beta)} \cdot W_m$$

$$\frac{W_T}{W_m} = \frac{\cos \phi}{\cos \beta \cos \phi \cdot \cos \beta (1 + \tan \phi \cdot \tan \beta)}$$

$$W_T = \frac{1 + \tan^2 \beta}{1 + \tan \phi \tan \beta} \cdot W_m$$

$$W_T = \frac{1}{1 + \tan \phi \tan \beta} \cdot W_m$$

$$W_m - W_T = W_T (\tan \phi \tan \beta)$$

$$W_m - W_T = VI \cos \phi \cdot \frac{\sin \phi}{\cos \phi} \cdot \sin \beta$$

$$\text{Error} = W_m - W_T = VI \sin \phi \cdot \tan \beta$$

$$\text{error} = \frac{W_m - W_T}{W_T} \times 100$$

$$= \frac{VI \sin \phi \cdot \tan \beta}{VI \cos \phi} \times 100$$

$$\% \text{ error} = \pm (\tan \phi \tan \beta \times 100)$$

Example 12:

A voltage $100 \sin \omega t + 40 \cos (3 \omega t - 30^\circ) + 50 \sin (5 \omega t + 45^\circ)$ V is applied to the pressure coil circuit of a wattmeter and through the current coil is passed a current of $8 \sin \omega t + 6 \cos (5 \omega t - 120^\circ)$ A. What percentage of reading on the wattmeter is due to fundamental?

- (a) 31.9% (b) 50%
(c) 91.2% (d) 156.7%

Ans:(c)

Sol: The wattmeter reading is

$$W = \frac{1}{2} [100 \times 8 \cos 0 + (6 \times 50) \sin (45 - (-30))] \\ = 438.82 \text{ W}$$

The fundamental power is

$$= \frac{1}{2} \times 100 \times 8 \Rightarrow 400 \text{ W}$$

The wattmeter reading w.r.t fundamental

$$\text{power} = \frac{400}{438.82} \times 100 \Rightarrow 91.2 \%$$

Example 13:

If the Reactance of the pressure coil circuit of a wattmeter is 1% of its resistance, calculate the % error due to this cause at power factor of 0.8 respectively.

- (a) 0.72% (b) 0.75%
(c) 0.71% (d) 0.78%

Ans: (b)

Sol: From Δ le $\tan \beta = \frac{X_p}{r_p}$

$$= \frac{0.01 \times r_p}{r_p} = 0.01$$

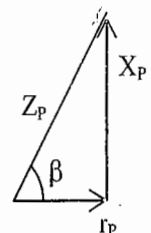
$$\beta = \tan^{-1}(0.01)$$

$$\cos \phi = 0.8$$

$$\phi = \cos^{-1}(0.8)$$

$$\tan \phi = \tan [\cos^{-1}(0.8)] = 0.75$$

$$\therefore \% \text{ error} = + (\tan \phi \times \tan \beta \times 100) \\ = + (0.75 \times 0.01 \times 100) \\ = 0.75 \%$$





⇒ Because of potential coil self inductance wattmeter reads more than the actual power for 'Lag' power factor error is +ve (for lag p.f)

⇒ Wattmeter reads less than the actual power and error is -ve for lead p.f

$$\text{Error} \propto \frac{1}{\cos \phi} \text{ (Lead/lag)}$$

Error Due to different load p.f

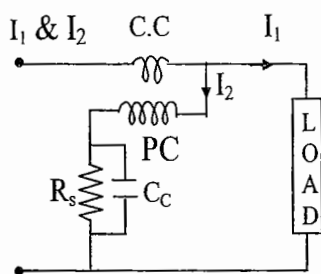
ϕ	$\cos \phi$	$\sin \phi$	$(W_m - W_T) = V.I. \sin \phi \cdot \tan \beta$
0	1	0	0 (min error)
30	0.866	0.5	$0.5 V.I \tan \beta$
90	0(low)	1	$VI \tan \beta$ (max error)

Note: Under low p.f load conditions the error in the measurement of the power is maximum in a normal (or) unity p.f. meter.

To reduce this error under low p.f load conditions. A specially designed low p.f watt meter is used.

Note: In case of O.C test of a transformer the p.f of transformer is between (0.2 to 0.4) under this low p.f condition to measure the power accurately L.P.F. watt meters are used.

Low power factor wattmeter: (Diagram of LPF)



In designing of the L.P.F. wattmeter

L.P.F. → $\cos \phi$ → low

$$T_d \propto I_1 I_2 \cos \phi$$

$$T_d \propto \frac{V}{R_s} I_1 \cdot \cos \phi$$

For L.P.F ' $\cos \phi$ ' is low. so that " T_d " is low to increase the value of " T_d " A series multiplier resistance ' R_s ' is kept low Value $R_s \downarrow$, $T_d \uparrow$

- To compensate Inductance of the 'PC' a Compensating capacitor is connected in parallel is

$$C_c \approx \frac{L_p}{R_s^2}$$

Note: In dynamometer watt meter the P.C should be highly resistive nature.

Example 14:

An electrodynamicometer wattmeter is used for measurement of power in a single phase circuit. The load voltage is 100V and the load current is 9A at a lagging power factor of 0.1. The wattmeter voltage circuit has a resistance of 3000Ω and inductance of 30mH Estimate the percent error in wattmeter reading when the pressure coil is connected on the supply side. The frequency is 60Hz.

- (a) 6.82% (b) 5.485%
(c) 9.82% (d) 7.44%

Ans: (d)

Sol: Load power = $100 \times 9 \times 0.1 = 90W$

Load power factor $\cos \phi = 0.1$

∴ $\phi = 84.26^\circ$ and $\sin \phi = 0.995$ and $\tan \phi = 9.95$

Resistance of pressure coil circuit $R_p =$

3000Ω

Reactance of pressure coil circuit =

$$2\pi \times 60 \times 30 \times 10^{-3} = 11.304\Omega$$



As the phase angle of the pressure angle of the pressure coil circuit is small

$$\beta = \tan \beta \approx \frac{11.304}{3000} = 0.00376 \text{ rad}$$

When the pressure coil is connected on the load side, the wattmeter measures power loss in pressure coil circuit in addition to load power.

$$\begin{aligned} \text{True power} &= VI \cos \phi \\ &= 100 \times 9 \times 0.1 \\ &= 90 \text{ W} \end{aligned}$$

Let us consider only the effect of Inductance
Reading of watt meter

$$\begin{aligned} &= \text{true power} (1 + \tan \phi \tan \beta) \\ &= 90 (1 + 9.95 \times 0.00376) \\ &= 93.36 \text{ W} \end{aligned}$$

$$\begin{aligned} \text{Power loss in pressure coil circuit} &= \frac{V^2}{R_p} \\ &= \frac{(100)^2}{3000} \\ &= 3.33 \text{ W} \end{aligned}$$

∴ Reading of wattmeter considering the power loss in pressure coil circuit.

$$= 93.36 \text{ W} + 3.33 = 96.697$$

$$\% \text{error} = \frac{96.69 - 90}{90} \times 100 = 7.44\%$$

Example 15:

The torque produced in an indicating instrument by utilizing magnetic, electro dynamic, thermal, chemical and electro static effect is known as.

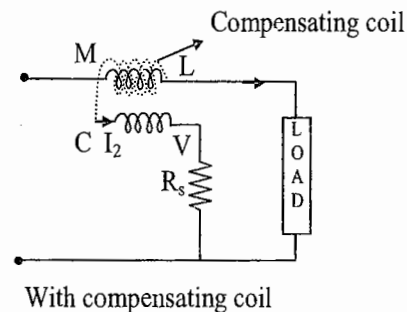
- Controlling torque
- Deflecting torque
- Damping torque
- Restoring torque

Ans:(b)

Compensation Coil (LC connection)

- In L.C Connection Current Coil carrying additionally potential coil current, because of this additional flux is produced in current coil & it links with P.C induces an EMF & causes current in potential coil because of this power loss is present in P.C
- To compensate the power loss in pressure coil compensating coil with equal number of turns of current coil is connected in series opposition to potential coil.

Compensating coil requires only for LC connection not for MC connection.



Errors in Dynamometer Type Instruments

1. Frictional Error (Torque/weight):

- Potential coil has more number of turns then weight of moving coil is increased.
- Then It has more Frictional error compared to M.C and M.I

2 Temperature Error:

- Because presence of two coils net heat developed is more. Then temperature error is more compare to other Instruments.



3 Frequency error:

- When it is used as a voltmeter (or) ammeter Net Inductance is more then the frequency error is more
- But when used as a wattmeter Inductance is very less, then frequency error is low.

Hystersis & Eddy current error:

- Because of the use of Air core, The hysteresis and eddy current errors are almost zero.
- Both Fixed coil & moving coils are electro magnet then poor magnetic field is present

- The external magnetic field can easily distort the inside operating field. Hence stray magnetic field is more compared to other

Note: For portable wattmeter to reduce the stray magnetic field error; Iron shielding is used.

- For precision watt meters to reduce stray magnetic field error, static system is used, in this two set of fixed coil and moving coils are used.